

Ethical Analysis of AI-Driven Art Generation Platform Artify

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Abstract

This essay conducts a systematic ethical analysis of Artify, an AI-driven art generation platform, focusing on its core ethical dilemmas of cultural bias and privacy risks through the lenses of utilitarianism, deontology, and virtue ethics. Each framework offers a distinct perspective on evaluating the platform's ethical flaws. Based on this multi-dimensional analysis, an integrated mitigation scheme is proposed, which combines the core strengths of the three frameworks to put forward practical, operable strategies for addressing Artify's ethical issues.

Ethical Analysis of Artify Based on Three Frameworks

1. Utilitarian Ethics Analysis

Definition: Utilitarian ethics takes the greatest happiness principle as the core, judging the morality of an act by the overall happiness and benefit it brings to the majority of people, and measuring the ethicality of a behavior by weighing the total positive and negative consequences.

For Artify platform, its creation has both obvious positive utility and potential negative utility.

In terms of positive impact, Artify makes artistic creation more accessible to the general public: even ordinary users without professional art training can generate artworks with their own ideas by inputting a few simple parameters. This virtually breaks down the professional barriers to artistic creation, satisfies the public's need for artistic expression, and brings emotional and spiritual fulfillment to a large number of ordinary people. At the same time, this platform has significantly driven the development of the emerging digital art industry, innovating artistic creation forms with the assistance of artificial intelligence and generating substantial economic and industrial benefits.

From a negative perspective, the bias in the training datasets leads to greater homogenization in artistic expression, marginalizing non-Western art forms and

undoubtedly harming the interests of these ethnic and cultural groups. This exposes artists from minority backgrounds to the risk of misappropriation and suppression of their work, causing them emotional distress. Furthermore, privacy and copyright disputes arising from AI-generated works may trigger numerous legal conflicts, causing not only psychological anxiety for users but also potential economic risks such as more severe financial losses. This reduces the overall economic utility for society in this area.

2. Deontological Ethics Analysis:

Definition: Deontological ethics is a rule-based ethical theory, which judges the morality of an act by whether the act itself abides by moral norms and moral duties, regardless of the consequences of the act. Its core is moral obligation and universal moral rules, emphasizing that people or organizations should abide by inherent moral responsibilities in their behaviors, such as respect for others, protection of rights, and abidance by justice.

From a deontological perspective, we find significant ethical risks associated with Artify: ①First, the platform has failed to fulfill its moral obligation to respect cultural diversity. Given its global reach, respecting different artistic traditions and preventing cultural bias in AI models due to flawed training datasets is paramount. This obligation stems from a fundamental moral requirement: fair treatment of all cultural forms.②Second, the platform's vague explanations regarding data collection and use violate users' right to know and fail to fulfill its privacy protection obligations. The platform has a moral responsibility to fully inform users about the scope, purpose, and sharing methods of data collection, ensuring their right to know and right to choose.③Third, the potential copyright infringement issues arising from AI-generated works on the platform indicate a failure to protect the intellectual property rights of original artists. As a platform closely linked to human-created works, it must adhere to its obligation to protect intellectual property rights and prevent AI-generated works from infringing on the copyrights of existing human works.

3. Virtue Ethics Analysis:

Definition: Virtue ethics focuses on the moral character and virtue cultivation of the subject, judging the morality of an act by whether the actor's character and motivation conform to the excellent moral virtues, and paying attention to the cultivation of the actor's stable moral quality rather than just the act itself or its consequences.

For Artify, we want to know whether the moral values cultivated by this platform are in line with the ethical standards of the artificial intelligence industry and the creative fields.

From the perspective of virtue ethics, Artify's startup has obvious defects in moral virtue cultivation: the motive of the enterprise is more inclined to pursuit of commercial benefits and technological innovation speed, while ignoring the cultivation of core virtues such as justice and respect. In the construction of the training dataset, the enterprise takes well-known artists and mainstream Western art works as the main content for the convenience of data collection and to meet the aesthetic needs of the majority of users, which is a manifestation of the lack of justice and respect virtues; in the data collection link, the enterprise fails to fully inform users of the use of data, which is a lack of honesty virtue; after the emergence of bias and privacy problems, the enterprise does not take active and effective measures to solve them, which is a lack of responsibility virtue. The lack of these core virtues makes the platform's operation behavior deviate from the ethical track of virtue ethics.

Integrated Mitigation Scheme

Solutions to mitigate ethical issues related to artificial intelligence should not be limited to a single framework, but should integrate the core strengths of three major frameworks and propose multi-dimensional solutions based on reality. This solution takes "adhering to moral rules (deontology) as the basic principle, balancing the rights and interests of multiple stakeholders (utilitarianism) as the core direction, and cultivating moral virtues of enterprises and users (virtue ethics) as the action method," and proposes specific implementation strategies to address Artify's bias and privacy issues.

1. Mitigation Strategies for Cultural Bias Problems

The system of datasets needs to be more rigorous and standardized (Deontology + Utilitarianism):

Based on the deontological requirement of respecting cultural and moral obligations, Artify should establish unified and standardized rules for constructing datasets, making these a mandatory baseline for algorithm development. These rules should clearly define the proportion of artworks from different cultures, regions, and eras in the training dataset, specifically increasing the weight of non-Western art forms, minority artistic traditions, and works by underrepresented artists. For example, the platform could collaborate with cultural museums and ethnic cultural research institutions in various countries to collect authentic and systematic non-Western artworks (such as Chinese ink painting, folk art from Africa and Latin America), ensuring that non-Western artworks account for no less than 50% of the total dataset. Simultaneously, a professional cultural review team, composed as many as possible of cultural researchers from diverse backgrounds, should be established to review the collected artworks, avoiding misrepresentation of cultural connotations in the dataset. From a utilitarian perspective, this measure can reduce the negative

effects of cultural marginalization, protect the interests of minority cultural creators, balance the distribution of utility among different cultural groups, and thus enhance the platform's overall social utility.

Encouraging the creation of artworks from minority backgrounds (Virtue):

To demonstrate its values of respecting and protecting virtues, Artify should prioritize the protection and encouragement of lesser-known artists. Firstly, by establishing an incentive award for artists from these backgrounds, the platform can continuously invite artists from minority cultural backgrounds to create original works. With their permission, these works can be included in the core training dataset, providing reasonable financial rewards while enriching the platform's material library. Secondly, the platform should recommend works by these artists to users at an appropriate ratio, clearly crediting them. This not only provides users with a novel aesthetic experience and helps attract new users but also respects the creative achievements of these artists, significantly improving the platform's reputation. Furthermore, users from diverse cultures should be allowed to optimize the AI model, ensuring that the generated works accurately reflect cultural connotations. For example, for users wanting to generate traditional Chinese landscape paintings, the AI model can accurately use famous Chinese landmarks and historical sites as background materials, avoiding the inclusion of Western landmarks or distinctive architecture, further enhancing the platform's positive impact.

2. Mitigation Strategies for Privacy and Copyright Problems

Improve the User Information Authorization and Protection System (Deontology + Utilitarianism):

Based on the duty theory's obligation to protect user rights, Artify should comprehensively improve its user information disclosure and authorization system. The platform adopts a visual information disclosure approach: collected user data is divided into essential data (such as usernames and account information) and personalized recommendation data (such as art preferences and social media information). The purpose of collection, scope of use, and retention time for each type of data are clearly marked on the user registration page using images and short texts. Simultaneously, the platform employs an independent authorization model, allowing users to modify their authorization scope at any time in their personal center. Based on the utilitarian goal of reducing the negative effects of data breaches and the duty theory's obligation to protect user privacy, Artify should build a data security protection system to safeguard user data security from both technical and institutional perspectives. The platform should establish a dedicated data security team, employing advanced data encryption technologies (such as end-to-end encryption and blockchain data storage) to encrypt users' personal data and art preference data. The platform's data system should be regularly tested and repaired to promptly address potential security vulnerabilities. Furthermore, the scope of access to privacy data should be strictly limited, stipulating that only the

professional technical team responsible for algorithm optimization can access user data, and the access process must be recorded in real time. This measure can effectively reduce the potential economic and psychological losses suffered by users due to data breaches, reduce the platform's negative utility (utilitarianism), and at the same time fulfill the company's moral obligation to protect user privacy (deontological ethics).

Establish an AI Art Copyright Resolution Mechanism (Deontology + Virtue Ethics):

To address copyright issues related to AI-generated works, Artify should establish a professional copyright dispute resolution mechanism. Firstly, it should formulate clear rules for determining the copyright of AI-generated art (obligatory approach). Combining AI technology with human review, the generated works should be compared with existing works in the training dataset, and similarity should be marked. If the similarity exceeds a preset threshold (e.g., 70%), the platform will refuse to generate the work and prompt the user to adjust parameters to avoid copyright infringement. Simultaneously, copyright protection should be an important part of the company culture (virtue ethics), cultivating a sense of responsibility and prudence within the company. The platform should regularly conduct copyright law training for its technology and operations teams, establish a copyright risk early warning system, and anticipate and mitigate copyright risks associated with AI-generated works. Before users generate works, the platform should clearly inform them of the ownership of the work (user ownership + platform technology copyright), the scope of use, and potential copyright risks through a pop-up window, and sign a concise and clear copyright agreement with the user.

Conclusion

The bias and privacy ethical dilemmas faced by the Artify platform are typical manifestations of ethical issues in the development of AI-powered creative technologies. Analysis based on utilitarianism, deontology, and virtue ethics shows that the ethical problems of AI technologies in the creative field are multidimensional and cannot be solved by a single ethical framework. This paper proposes an integrated mitigation solution that combines the advantages of these three ethical frameworks, offering specific and actionable strategies to address Artify's bias and privacy issues.

For the development of AI technologies in the creative field, ethical construction is not a constraint on technological innovation, but a necessary guarantee for sustainable technological development. Only by integrating ethical considerations into the entire process of AI research, development, operation, and supervision, adhering to moral obligations, balancing the rights and interests of all stakeholders, and cultivating ethical virtues in technology companies can AI truly democratize artistic creation, promote the diversified development of the creative field, and bring greater positive value to society as a whole.